SHRI GOVINDRAM SEKSARIA INSTITUTE OF TECHNOLOGY AND SCIENCE INDORE (M.P)

Department of Computer Technology and Applications

MCA First Year

Semester - II

Session 2025-26

Subject: - Operating System

LAB ASSIGNMENT - II

Submitted To: -

Prof. Priyanka Kabra
Prof. Shweta Gupta

Submitted By:

VaibhavPanchal
(0801CA251144)

INDEX

S.No.	Assignment Questions	Page No.	Signature
1	**<u>Creating a file and a file name students.txt contains student records in the following format:</u>** RollNo:Name:Department:Marks 101:Rahul:CSE:78 102:Asha:IT:85 103:Ravi:CSE:65 104:Neha:ECE:90 105:Karan:CSE:78 Tasks : ◆ Display all records of CSE students. ◆ Count the total number of students. ◆ Display names of students who scored more than 75 marks. ◆ Sort the file based on marks (descending order). ◆ Find students having duplicate marks. **Using the grep, wc, awk, sort, uniq commands for this task**	1	
2	**<u>Case Study : Payroll Processing</u>** Creating an employee salary file name salary.txt and contains the following records in the following format: EmpID Name Department Salary E101 Amit HR 30000 E102 Rina CTA 45000 E103 Suresh CTA 50000 E104 Meena HR 35000	4	

	Perform the following task : ◆ Display employees belonging to the IT department. ◆ Print only Name and Salary columns. ◆ Increase salary by 10% for IT employees. ◆ Calculate the average salary. ◆ Count employees in each department. **Using the awk, cut, grep commands.**		
3	**<u>Case study :Automated Monitoring Report</u>** ◆ A daily monitoring report is required for system health. TasksCollect CPU, memory, and disk usage information. ◆ Save the report with date and time. ◆ Schedule the report generation automatically. ◆ View historical monitoring data. **Using top, free, df, date, cron, sar for complete these tasks.**	**7**	
4	**<u>Case Study :File Copy, Move, and Rename</u>** **Scenario** **Project files need to be organized into appropriate directories.** **<u>Tasks:</u>** ◆ Create a directory project. ◆ Copy report.txt into the project directory. ◆ Rename report.txt to final_report.txt. ◆ Move final_report.txt to a directory named backup. ◆ Verify the file location. **Using mkdir, cp, mv, ls commands for this task.**	**10**	
	<u>Case Study :File Deletion and Recovery</u>		

5	**<u>Awareness</u>** **Scenario** **Old and unused files must be removed from the system.** **Tasks:** ◆ Delete a file named old.txt. ◆ Remove an empty directory. ◆ Remove a directory containing files. ◆ List files before and after deletion. ◆ Explain why deleted files cannot be easily recovered. **Using rm, rmdir, ls for this task.**	**13**	
6	**<u>Check Current User and Group:</u>** Open a terminal and run the command to check your current user and group. **Setting Up the Environment:** ◆ Create a directory named file_permission_lab. ◆ Inside this directory, create three files: file1.txt, file2.txt, and file3.txt.	**15**	
7	**<u>Viewing File Permissions:</u>** ◆ Use the command to view the permissions of the files you created. Document the output.	**17**	
8	**<u>Changing File Permissions:</u>** ◆ Change the permissions of file1.txt to allow the owner to read and write, the group to read, and others to have no permissions. Use both symbolic and numeric methods. ◆ Change the permissions of file2.txt to allow everyone to read and execute but not write. ◆ Change the permissions of file3.txt to	**18**	

	allow only the owner to read, write, and execute.		
9	**<u>Documenting Changes:</u>** ◆ After each permission change, use ls -l to document the new permissions for each file.	**20**	
10	**<u>File Handling Tasks:</u>** **1.** Create a script called file_handling.sh that: • Creates a directory named test_dir. • Creates a file named test_file.txt inside test_dir. • Writes "This is a test file." to the file. • Displays the contents of the file. • Deletes the file and the directory. **2.** Make the script executable and run it with different arguments.	**21**	
11	**<u>Change Ownership:</u>** ◆ If you have another user account on the system, change the ownership of file1.txt to that user (replace other user with the actual username)	**22**	
12	**<u>Cleanup:</u>** ◆ After completing the tasks, remove the file_permission_lab directory and its contents	**23**	
13	**<u>File commands:</u>** **Create a Directory:** ◆ Create a new directory named file_commands_lab in your home directory. **Navigate to the Directory:** ◆ Change to the newly created directory ◆ Create five text files named file1.txt, file2.txt, file3.txt, file4.txt, and file5.txt.	**24**	

	◆ Use the command to list the files in the directory.		
14	**<u>Write to Files:</u>** ◆ Use the echo command to write “This is file 1” text into file1.txt on the terminal not in the directly writing text file. ◆ Append text to file2.txt.write the contents "This is file 2” in the file2.txt and that append.	**26**	
15	◆ Use the cat command to display the contents of file1.txt. ◆ Use the less command to view the contents of ‘file2.txt’	**27**	
16	**<u>Copy Files:</u>** ◆ Copy ‘file1.txt’ to create a new file named ‘file1_copy.txt’ ◆ Rename ‘file2.txt ‘ to ‘file2_renamed.txt’ ◆ Delete ‘file3.txt’. ◆ Change to the ‘backup directory and list the files:	**28**	
17	**<u>Create a Compressed Archive:</u>** ◆ Go back to the ‘file_commands_lab’ directory and create a compressed archive of the backup directory.	**30**	
18	**<u>Extract the Archive:</u>** ◆ Create a new directory named ‘extracted_backup’ and extract the contents of ‘backup.tar.gz’ into it:	**31**	
19	**<u>Verify Extraction:</u>** ◆ List the contents of the ‘extracted_backup’ directory to verify that the files were extracted correctly.	**32**	
20	**Create a text file named “ count.txt “ and write the 10 line sentences and use the command to count the word ,lines , character and also use command for highlighting a patten in this file .**	**33**	

Question 1:

<u>Creating a file and a file name students.txt contains student records in the following format:</u>

RollNo:Name:Department:Marks
101:Rahul:CSE:78
102:Asha:IT:85
103:Ravi:CSE:65
104:Neha:ECE:90
105:Karan:CSE:78

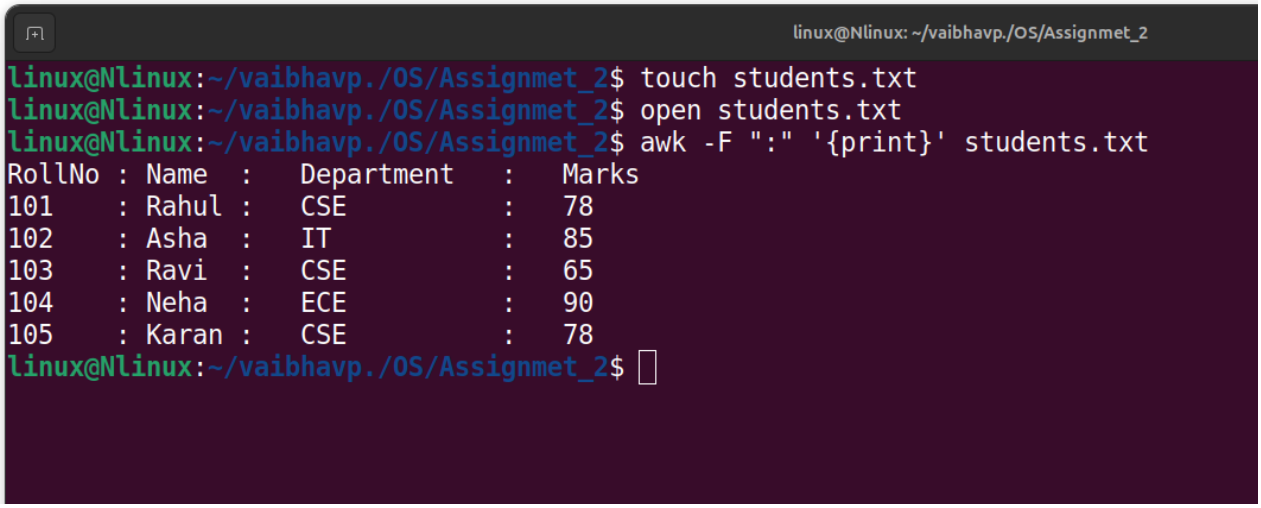
Tasks :

- **Display all records of CSE students.**
- **Count the total number of students.**
- **Display names of students who scored more than 75 marks.**
- **Sort the file based on marks (descending order).**
- **Find students having duplicate marks.**

Using the grep, wc, awk, sort, uniq commands for this task

<u>Commands:</u>

linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ touch students.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ open students.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ awk -F ":" '{print}' students.txt



1. **Display all records of CSE students.**

Command:
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ grep "CSE" students.txt

```
linux@Nlinux: ~/vaibhavp./OS/Assignmet_2
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ grep "CSE" students.txt
101     : Rahul :    CSE          :    78
103     : Ravi  :    CSE          :    65
105     : Karan :    CSE          :    78
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$
```

2. **Count the total number of students.**

Command:
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ wc -l students.txt

```
linux@Nlinux: ~/vaibhavp./OS/Assignmet_2
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ wc -l students.txt
6 students.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$
```

3. **Display names of students who scored more than 75 marks.**

Command:
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ awk -F: '$4 > 75 {print $2}' students.txt

```
linux@Nlinux: ~/vaibhavp./OS/Assignmet_2
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ awk -F: '$4 > 75 {print $2}' students.txt
 Rahul
 Asha
 Neha
 Karan
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$
```

4. **Sort the file based on marks (descending order).**

Command:
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ sort -t: -k4 -nr students.txt

```
linux@Nlinux: ~/vaibhavp./OS/Assignmet_2
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ sort -t: -k4 -nr students.txt
104     : Neha  :   ECE          :   90
102     : Asha  :   IT           :   85
105     : Karan :   CSE          :   78
101     : Rahul :   CSE          :   78
103     : Ravi  :   CSE          :   65
RollNo  : Name  :   Department   :   Marks
```

5. **Find students having duplicate marks.**

Command:

linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ awk -F: '{print $4 ":" $0}' students.txt | sort | uniq -d -w2 | cut -d: -f2-

```
linux@Nlinux: ~/vaibhavp./OS/Assignmet_2
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ awk -F: '{print $4 ":" $0}' students.txt | sort | uniq -d -w2 | cut -d: -f2-
103     : Ravi  :   CSE          :   65
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$
```

Question 2:

<u>Case Study : Payroll Processing</u>

Creating an employee salary file name salary.txt and contains the following records in the following format:

EmpID Name Department Salary
E101 Amit HR 30000
E102 Rina CTA 45000
E103 Suresh CTA 50000
E104 Meena HR 35000

Perform the following task :

1. **Display employees belonging to the IT department.**
2. **Print only Name and Salary columns.**
3. **Increase salary by 10% for IT employees.**
4. **Calculate the average salary.**
5. **Count employees in each department.**

Using the awk, cut, grep commands.

Commands:

linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ touch salary.csv
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ libreoffice salary.csv

	A	B	C	D
1	EmpID	Name	Department	Salary
2	E101	Amit	HR	30000
3	E102	Rina	CTA	45000
4	E103	Suresh	CTA	50000
5	E104	Meena	HR	35000
6	E105	Ram	IT	55000
7	E106	Shyam	IT	60000

1. **Display employees belonging to the IT department.**

Command:

linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ grep "IT" salary.csv

```
linux@Nlinux: ~/vaibhavp./OS/Assignmet_2
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$  grep "IT" salary.csv
E105,Ram,IT,55000
E106,Shyam,IT,60000
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$
```

2. **Print only Name and Salary columns.**

Command:

linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ awk -F, '{print $2,$4}' salary.csv

```
linux@Nlinux: ~/vaibhavp./OS/Assignmet_2
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ awk -F, '{print $2,$4}' salary.csv
Name Salary
Amit 30000
Rina 45000
Suresh 50000
Meena 35000
Ram 55000
Shyam 60000
```

3. **Increase salary by 10% for IT employees.**

Command:

linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ awk -F, '{if ($3=="IT") $4 = $4 * 1.10; print}' salary.csv

```
E105 Ram IT 60500
E106 Shyam IT 66000
```

4. **Calculate the average salary.**

Command:

linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ awk -F, 'NR>1 { sum += $4; count++ } END { print "Average Salary =", sum/count }' salary.csv

```
linux@Nlinux: ~/vaibhavp./OS/Assignmet_2
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ awk -F, 'NR>1 { sum += $4; count++ } END { print "Average Salary =", sum/count }' salary.csv
Average Salary = 45833.3
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$
```

5. **Count employees in each department.**

Command:

linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ awk -F, 'NR>1 { dept[$3]++ } END {for (d in dept) print d, dept[d]}' salary.csv

```
linux@Nlinux: ~/vaibhavp./OS/Assignmet_2
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ awk -F, 'NR>1 { dept[$3]++ } END {for (d in dept) print d, dept[d]}' salary.csv
IT 2
CTA 2
HR 2
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$
```

Question 3:

<u>Case study: Automated Monitoring Report</u>

A daily monitoring report is required for system health.

Tasks:

1. **Collect CPU, memory, and disk usage information.**
2. **Save the report with date and time.**
3. **Schedule the report generation automatically.**
4. **View historical monitoring data.**

Using top, free, df, date, cron, sar for complete these tasks.

1. **Collect CPU, memory, and disk usage information.**

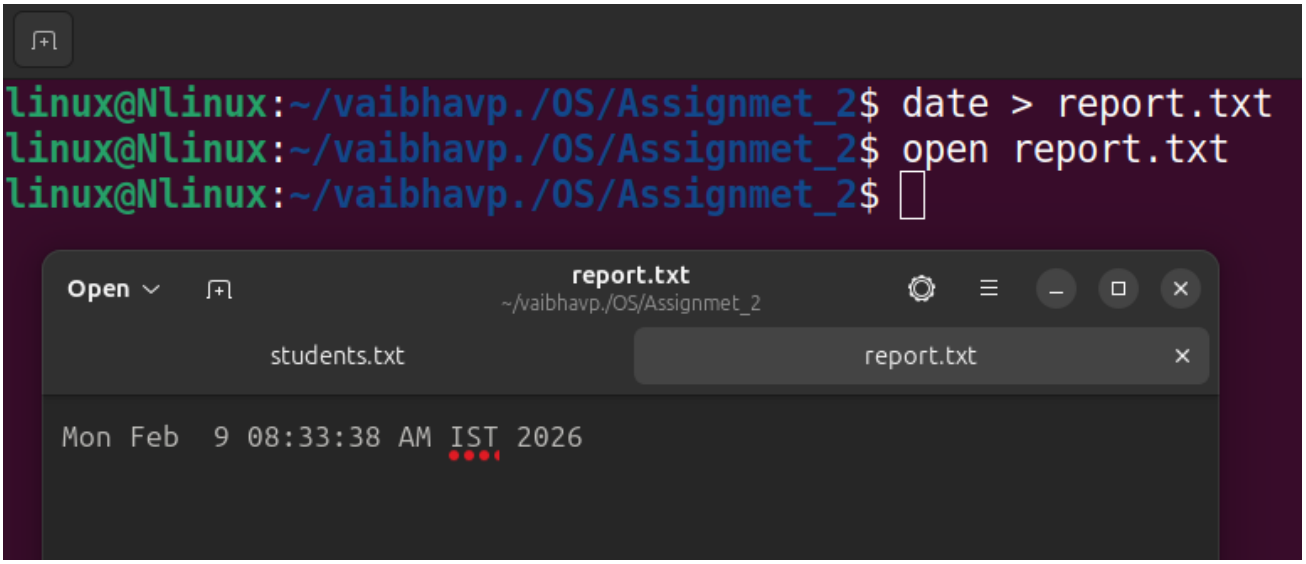
Commands:
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ top -b -n 1 | head -5
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ free -h
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ df -h

```
linux@Nlinux: ~/vaibhavp./OS/Assignmet_2
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ top -b -n 1 | head -5
top - 08:31:53 up  1:27,  1 user,  load average: 0.30, 0.56, 0.58
Tasks: 360 total,   1 running, 359 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.7 us,  1.5 sy,  0.0 ni, 97.8 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  15696.6 total,   7402.2 free,   5607.1 used,   4055.5 buff/cache
MiB Swap:   7813.0 total,   7813.0 free,      0.0 used.  10089.5 avail Mem
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ free -h
               total        used        free      shared  buff/cache   available
Mem:            15Gi       5.5Gi       7.2Gi       1.0Gi       4.0Gi       9.9Gi
Swap:          7.6Gi          0B       7.6Gi
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           1.6G  2.4M  1.6G   1% /run
/dev/nvme0n1p9   91G   18G   69G  21% /
tmpfs           7.7G   54M  7.7G   1% /dev/shm
tmpfs           5.0M   12K  5.0M   1% /run/lock
efivarfs        268K  190K   74K  72% /sys/firmware/efi/efivars
/dev/nvme0n1p7  944M  132M  747M  15% /boot
/dev/nvme0n1p1  196M   37M  160M  19% /boot/efi
tmpfs           1.6G  152K  1.6G   1% /run/user/1000
/dev/nvme0n1p6  276G  100G  177G  37% /media/linux/New Volume
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$
```

2. **Save the report with date and time.**

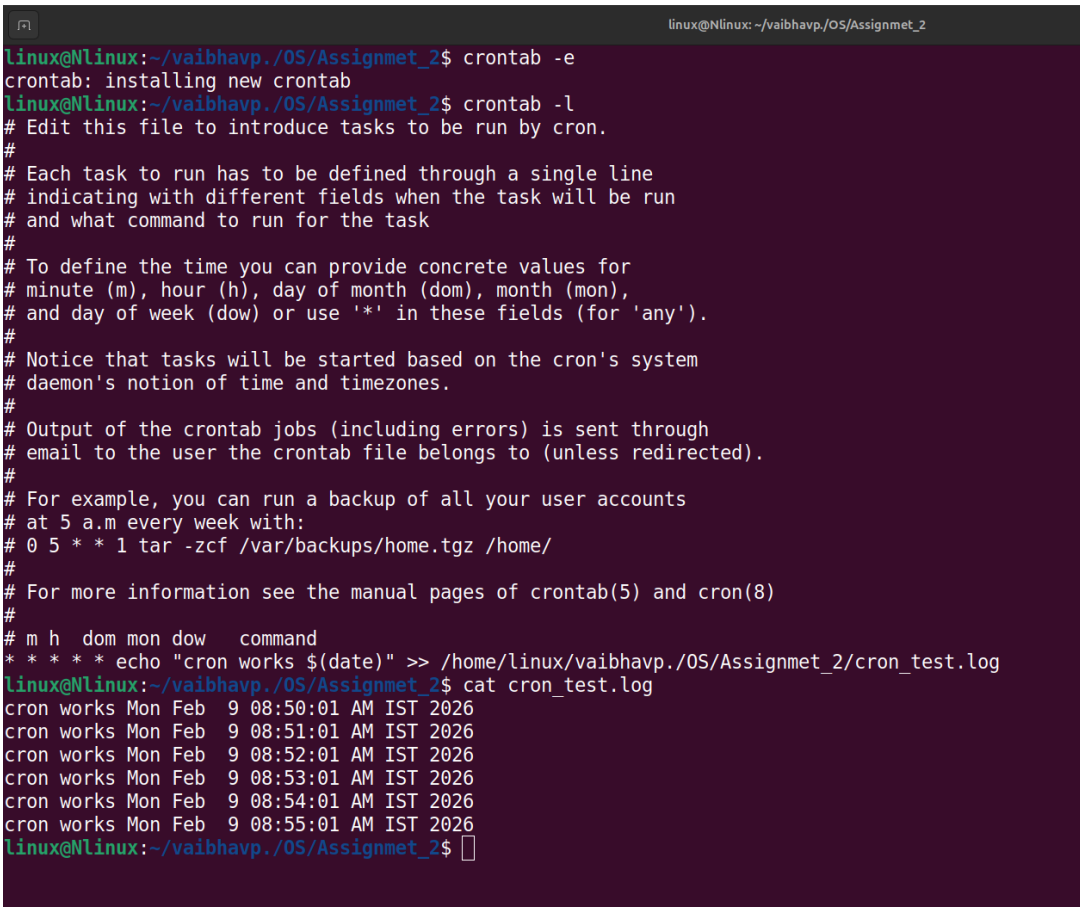
Commands:
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ date > report.txt

linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ open report.txt



3. **Schedule the report generation automatically.**

Commands:
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ crontab -e
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ crontab -l
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ cat cron_test.log



4. **View historical monitoring data.**

Commands:

linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ sar

```
linux@Nlinux: ~/vaibhavp./OS/Assignmet_2

linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ sar
Linux 6.14.0-37-generic (Nlinux)        02/09/2026      _x86_64_        (12 CPU)

01:15:31 AM  LINUX RESTART      (12 CPU)

01:20:02 AM     CPU     %user     %nice   %system   %iowait    %steal     %idle
01:30:11 AM     all      3.65      0.01      1.03      0.14      0.00     95.17
01:40:08 AM     all      4.73      0.01      1.15      0.16      0.00     93.95
01:50:00 AM     all      4.92      0.00      1.20      0.14      0.00     93.74
02:00:11 AM     all      6.88      0.00      1.82      0.75      0.00     90.55
02:10:11 AM     all      7.18      0.00      1.81      0.49      0.00     90.51
02:20:02 AM     all      4.30      0.00      0.95      0.22      0.00     94.52
02:30:09 AM     all      4.35      0.01      1.00      0.21      0.00     94.44
02:40:04 AM     all      3.56      0.00      0.92      0.11      0.00     95.41
02:50:02 AM     all      7.29      0.01      1.66      0.50      0.00     90.54
03:00:02 AM     all     11.89      0.02      2.95      0.92      0.00     84.22
03:10:11 AM     all      4.77      0.00      1.08      0.17      0.00     93.97
03:20:02 AM     all      4.57      0.00      1.11      0.11      0.00     94.20
03:30:05 AM     all      5.29      0.00      1.20      0.14      0.00     93.37
03:40:11 AM     all      3.57      0.00      0.67      0.08      0.00     95.68
03:50:01 AM     all      5.29      0.00      0.98      0.11      0.00     93.63
04:00:01 AM     all      4.78      0.00      0.88      0.09      0.00     94.25
04:10:08 AM     all      4.97      0.00      0.91      0.10      0.00     94.03
04:20:02 AM     all      5.62      0.00      1.15      0.14      0.00     93.09
04:30:11 AM     all      5.47      0.00      1.19      0.11      0.00     93.23
04:40:11 AM     all      6.07      0.00      1.27      0.18      0.00     92.47
04:50:02 AM     all      5.71      0.00      1.29      0.19      0.00     92.81
05:00:11 AM     all      4.48      0.00      0.91      0.07      0.00     94.54
05:10:06 AM     all      3.81      0.00      0.91      0.09      0.00     95.20
05:20:02 AM     all      2.67      0.01      0.97      0.08      0.00     96.28
05:30:08 AM     all      4.54      0.00      1.17      0.31      0.00     93.98
Average:        all      5.21      0.00      1.21      0.22      0.00     93.36

07:04:41 AM  LINUX RESTART      (12 CPU)

07:10:01 AM     CPU     %user     %nice   %system   %iowait    %steal     %idle
07:20:13 AM     all      3.31      0.00      0.79      0.09      0.00     95.82
```

Question 4:

<u>Case Study :File Copy, Move, and Rename</u>

Scenario

Project files need to be organized into appropriate directories.

<u>Tasks:</u>

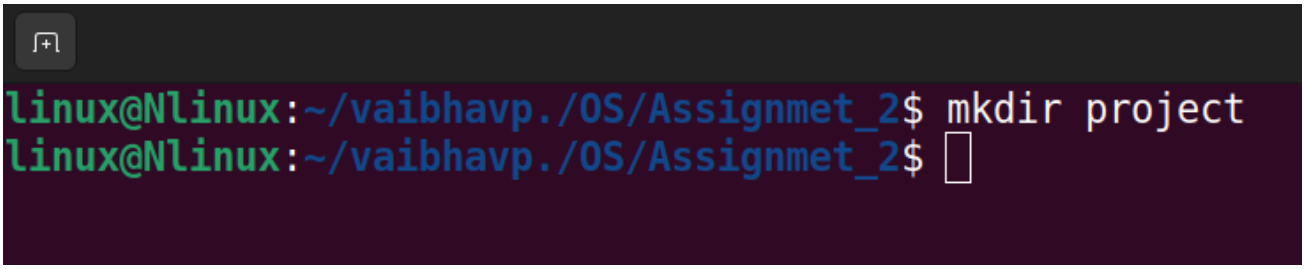
1. **Create a directory project.**
2. **Copy report.txt into the project directory.**
3. **Rename report.txt to final_report.txt.**
4. **Move final_report.txt to a directory named backup.**
5. **Verify the file location.**

Using mkdir, cp, mv, ls commands for this task.

1. **Create a directory project.**

Commands:

linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ mkdir project



2. **Copy report.txt into the project directory.**

Commands:

linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ cp report.txt project/

linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ ls project/

```
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$  cp report.txt project/
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ ls project/
report.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$
```

3. **Rename report.txt to final_report.txt.**

Commands:

linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ mv project/report.txt project/final_report.txt

linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ ls project/

```
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ mv project/report.txt project/final_report.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ ls project/
final_report.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$
```

4. **Move final_report.txt to a directory named backup.**

Commands:

linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ mkdir backup

linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ mv project/final_report.txt backup/

linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ ls backup/

```
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ mkdir backup
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ mv project/final_report.txt backup/
```

5. **Verify the file location.**

Commands:

linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ ls backup/

```
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ ls backup/
final_report.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$
```

Question 5:

<u>Case Study :File Deletion and Recovery Awareness</u>

<u>Scenario</u>
<u>Old and unused files must be removed from the system.</u>

<u>Tasks:</u>

1. **Delete a file named old.txt.**
2. **Remove an empty directory.**
3. **Remove a directory containing files.**
4. **List files before and after deletion.**
5. **Explain why deleted files cannot be easily recovered.**

<u>Using rm, rmdir, ls for this task.</u>

1. **Delete a file named old.txt.**

Commands:
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ touch old.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ rm old.txt

```
linux@Nlinux: ~/vaibhavp./OS/Assignmet_2
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ touch old.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ ls
backup  cron_test.log  new.csv  old.txt  project  report.sh  salary.csv  students.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ rm old.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ ls
backup  cron_test.log  new.csv  project  report.sh  salary.csv  students.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$
```

2. **Remove an empty directory.**

Commands:
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ rmdir empty/

```
linux@Nlinux: ~/vaibhavp./OS/Assignmet_2
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ mkdir empty
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ ls
backup  cron_test.log  empty  new.csv  project  report.sh  salary.csv  students.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ ls empty/
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ rmdir empty/
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ ls
backup  cron_test.log  new.csv  project  report.sh  salary.csv  students.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$
```

3. **Remove a directory containing files.**

Commands:
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ rm -r file/

```
linux@Nlinux: ~/vaibhavp./OS/Assignmet_2
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ mkdir file
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ mkdir file/new1
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ ls
backup  cron_test.log  file  new.csv  project  report.sh  salary.csv  students.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ ls file/
new1
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ touch file/new.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ ls file/
new1  new.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ rm -r file/
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ ls
backup  cron_test.log  new.csv  project  report.sh  salary.csv  students.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$
```

4. **List files before and after deletion.**

Commands:
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ ls
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ ls file/

```
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ ls
backup  cron_test.log  file  new.csv  project  report.sh  salary.csv  students.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ ls file/
new1
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ touch file/new.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ ls file/
new1  new.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ rm -r file/
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ ls
backup  cron_test.log  new.csv  project  report.sh  salary.csv  students.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$
```

5. **Explain why deleted files cannot be easily recovered.**

Explanation:
When a file is deleted using rm, the system removes its reference from the file system. The actual data may be overwritten later, making recovery difficult. Deleted files cannot be easily recovered primarily because when you delete a file, the operating system does not actually remove the data from the storage device. Instead, it deletes the file's index (its entry in the file system table) and marks the space it occupied as "available" for new data.

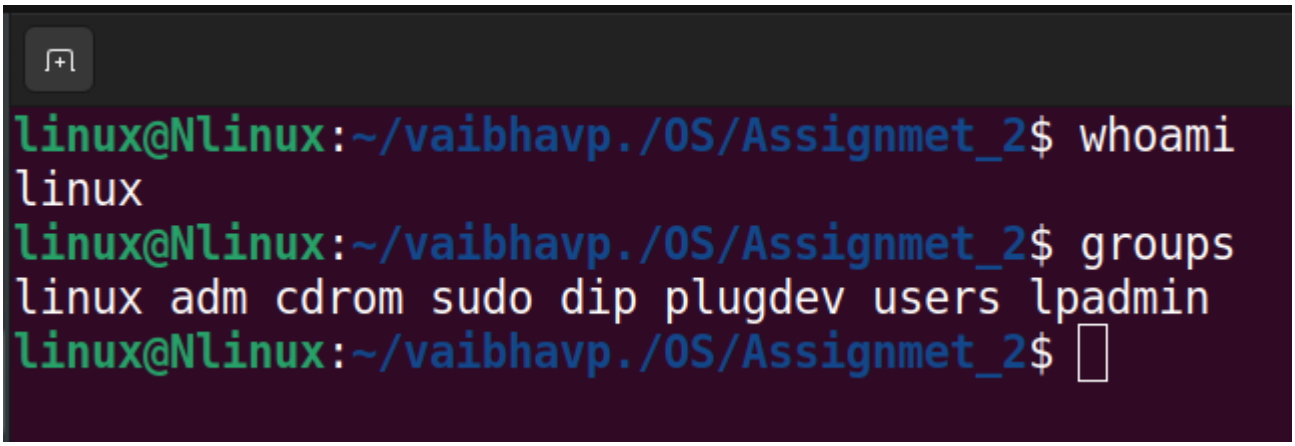
Question 6:

<u>Check Current User and Group:</u>
Open a terminal and run the command to check your current user and group.

Setting Up the Environment:

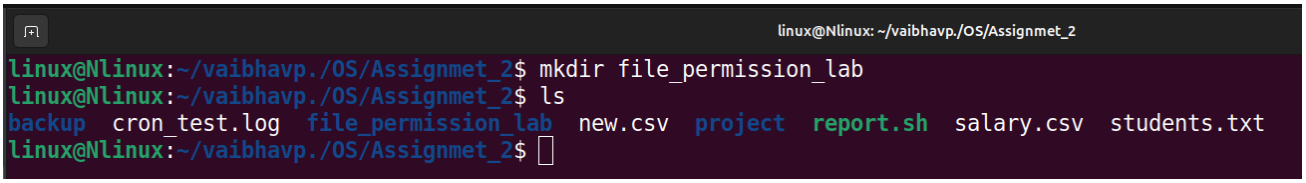
1. **Create a directory named file_permission_lab.**
2. **Inside this directory, create three files: file1.txt, file2.txt, and file3.txt.**

Commands:
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ whoami
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ groups



1. **Create a directory named file_permission_lab.**

Commands:
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ mkdir file_permission_lab



2. **Inside this directory, create three files: file1.txt, file2.txt, and file3.txt.**

Commands:

linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ cd file_permission_lab/

linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_permission_lab$ touch file1.txt file2.txt file3.txt

```
linux@Nlinux: ~/vaibhavp./OS/Assignmet_2/file_permission_lab
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ cd file_permission_lab/
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_permission_lab$ touch file1.txt file2.txt file3.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_permission_lab$ ls
file1.txt  file2.txt  file3.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_permission_lab$
```

Question 7:

<u>Viewing File Permissions:</u>

- **Use the command to view the permissions of the files you created. Document the output.**

Commands:

linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ ls -l

```
linux@Nlinux: ~/vaibhavp./OS/Assignmet_2
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ ls -l
total 72
drwxrwxr-x 2 linux linux  4096 Feb  9 09:02 backup
-rw-rw-r-- 1 linux linux 40463 Feb 16 01:15 cron_test.log
drwxrwxr-x 2 linux linux  4096 Feb  9 09:19 file_permission_lab
-rw-rw-r-- 1 linux linux   168 Feb  9 08:06 new.csv
drwxrwxr-x 2 linux linux  4096 Feb  9 09:02 project
-rwxrwxr-x 1 linux linux   264 Feb  9 08:40 report.sh
-rw-rw-r-- 1 linux linux   168 Feb  9 08:07 salary.csv
-rw-rw-r-- 1 linux linux   210 Feb  9 07:48 students.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$
```

Question 8:

<u>Changing File Permissions:</u>

- **Change the permissions of file1.txt to allow the owner to read and write, the group to read, and others to have no permissions. Use both symbolic and numeric methods.**

Commands:

linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_permission_lab$ ls -l
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_permission_lab$ chmod u=rw,g=r,o= file1.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_permission_lab$ ls –l

```
linux@Nlinux: ~/vaibhavp./OS/Assignmet_2/file_permission_lab
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_permission_lab$ ls -l
total 0
-rw-rw-r-- 1 linux linux 0 Feb  9 09:19 file1.txt
-rw-rw-r-- 1 linux linux 0 Feb  9 09:19 file2.txt
-rw-rw-r-- 1 linux linux 0 Feb  9 09:19 file3.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_permission_lab$ chmod u=rw,g=r,o= file1.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_permission_lab$ ls -l
total 0
-rw-r----- 1 linux linux 0 Feb  9 09:19 file1.txt
-rw-rw-r-- 1 linux linux 0 Feb  9 09:19 file2.txt
-rw-rw-r-- 1 linux linux 0 Feb  9 09:19 file3.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_permission_lab$ ~
```

- **Change the permissions of file2.txt to allow everyone to read and execute, but not write.**

Commands:

linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_permission_lab$ ls -l
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_permission_lab$ chmod a=rx file2.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_permission_lab$ ls –l

```
linux@Nlinux: ~/vaibhavp./OS/Assignmet_2/file_permission_lab
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_permission_lab$ ls -l
total 0
-rw-r----- 1 linux linux 0 Feb  9 09:19 file1.txt
-rw-rw-r-- 1 linux linux 0 Feb  9 09:19 file2.txt
-rw-rw-r-- 1 linux linux 0 Feb  9 09:19 file3.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_permission_lab$ chmod a=rx file2.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_permission_lab$ ls -l
total 0
-rw-r----- 1 linux linux 0 Feb  9 09:19 file1.txt
-r-xr-xr-x 1 linux linux 0 Feb  9 09:19 file2.txt
-rw-rw-r-- 1 linux linux 0 Feb  9 09:19 file3.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_permission_lab$
```

- **Change the permissions of file3.txt to allow only the owner to read, write, and execute.**

Commands:

linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_permission_lab$ ls -l
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_permission_lab$ chmod a=rx file2.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_permission_lab$ ls –l

```
linux@Nlinux: ~/vaibhavp./OS/Assignmet_2/file_permission_lab
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_permission_lab$ ls -l
total 0
-rw-r----- 1 linux linux 0 Feb  9 09:19 file1.txt
-r-xr-xr-x 1 linux linux 0 Feb  9 09:19 file2.txt
-rw-rw-r-- 1 linux linux 0 Feb  9 09:19 file3.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_permission_lab$ chmod u=rwx,g=,o= file3.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_permission_lab$ ls -l
total 0
-rw-r----- 1 linux linux 0 Feb  9 09:19 file1.txt
-r-xr-xr-x 1 linux linux 0 Feb  9 09:19 file2.txt
-rwx------ 1 linux linux 0 Feb  9 09:19 file3.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_permission_lab$
```

Question 9:

<u>Documenting Changes:</u>

- **After each permission change, use ls -l to document the new permissions for each file.**

Commands:
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_permission_lab$ ls -l

```
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_permission_lab$ ls -l
total 0
-rw-r----- 1 linux linux 0 Feb  9 09:19 file1.txt
-r-xr-xr-x 1 linux linux 0 Feb  9 09:19 file2.txt
-rwx------ 1 linux linux 0 Feb  9 09:19 file3.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_permission_lab$
```

Question 10:

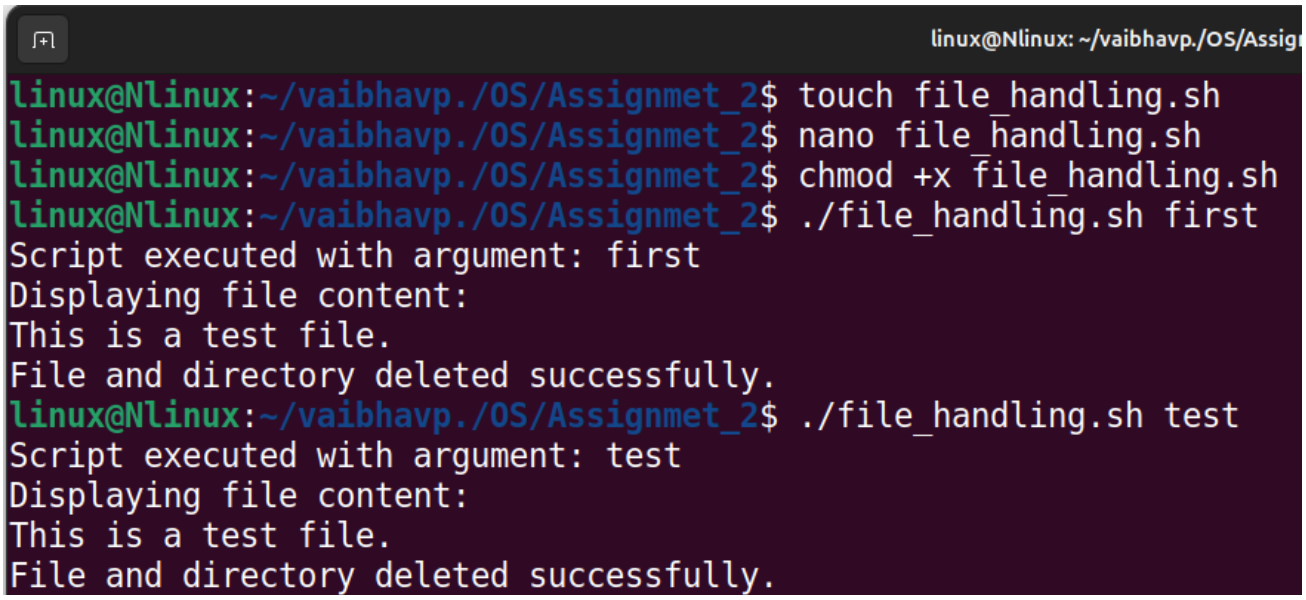
<u>File Handling Tasks:</u>

1. Create a script called file_handling.sh that:

- **Creates a directory named test_dir.**
- **Creates a file named test_file.txt inside test_dir.**
- **Writes "This is a test file." to the file.**
- **Displays the contents of the file.**
- **Deletes the file and the directory.**

2. Make the script executable and run it with different arguments.

Commands:
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ touch file_handling.sh
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ nano file_handling.sh
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ chmod +x file_handling.sh
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ ./file_handling.sh
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$./file_handling.sh test



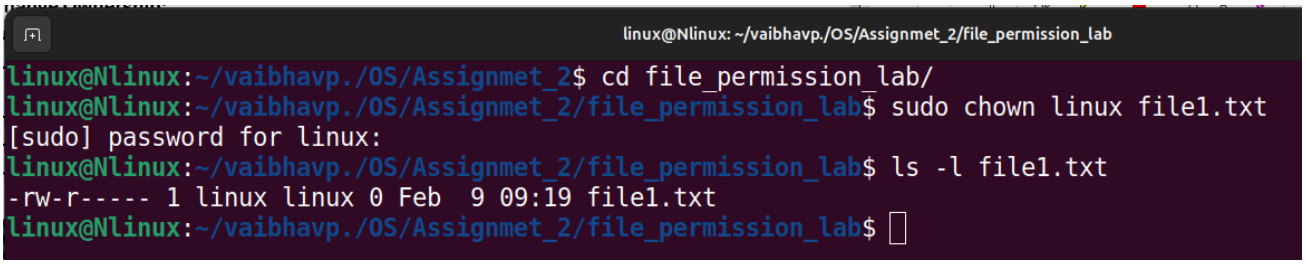
Question 11:

<u>Change Ownership:</u>

- **If you have another user account on the system, change the ownership of file1.txt to that user (replace other user with the actual username)**

Commands:

linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ cd file_permission_lab/
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_permission_lab$ sudo chown linux file1.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_permission_lab$ ls -l file1.txt



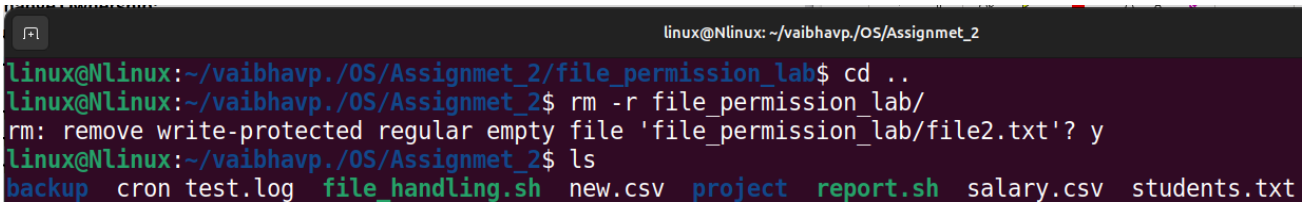
Question 12:

<u>Cleanup:</u>

- **After completing the tasks, remove the file_permission_lab directory and its contents.**

Commands:

linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ rm -r file_permission_lab/
linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ ls



Question 13:

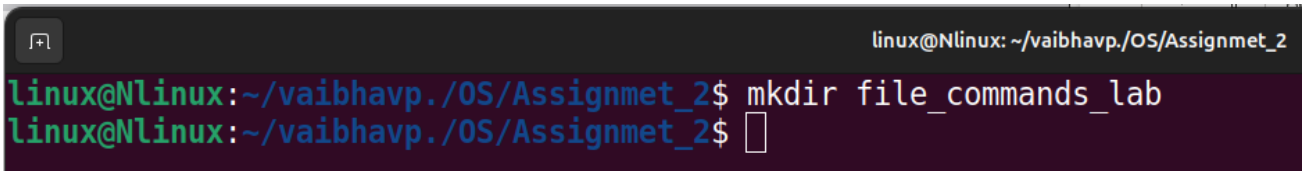
<u>File commands:</u>

Create a Directory:

- **Create a new directory named file_commands_lab in your home directory.**

Commands:

linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ mkdir file_commands_lab

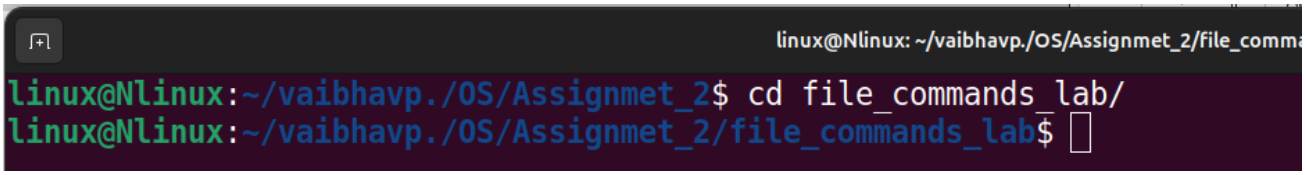


Navigate to the Directory:

- **Change to the newly created directory**

Commands:

linux@Nlinux:~/vaibhavp./OS/Assignmet_2$ cd file_commands_lab/



- **Create five text files named file1.txt, file2.txt, file3.txt, file4.txt, and file5.txt.**

Commands:

linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ touch file1.txt file2.txt file3.txt file4.txt file5.txt

```
linux@Nlinux: ~/vaibhavp./OS/Assignmet_2/file_commands_lab
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ touch file1.txt file2.txt file3.txt file4.txt file5.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$
```

- **Use the command to list the files in the directory.**

Commands:

linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ ls

```
linux@Nlinux: ~/vaibhavp./OS/Assig
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ ls
file1.txt  file2.txt  file3.txt  file4.txt  file5.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$
```

Question 14:

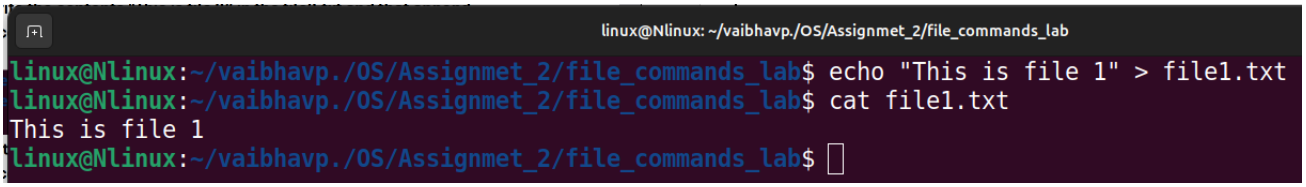
<u>Write to Files:</u>

- **Use the echo command to write “This is file 1” text into file1.txt on the terminal not in the directly writing text file.**

Commands:

linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ echo "This is file 1" > file1.txt

linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ cat file1.txt

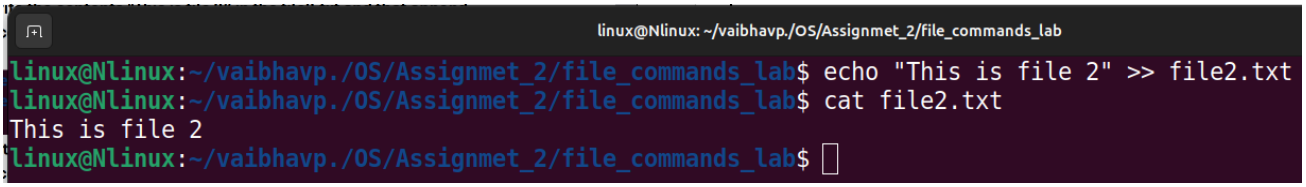


- **Append text to file2.txt.write the contents "This is file 2” in the file2.txt and that append.**

Commands:

linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ echo "This is file 2" >> file2.txt

linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ cat file2.txt

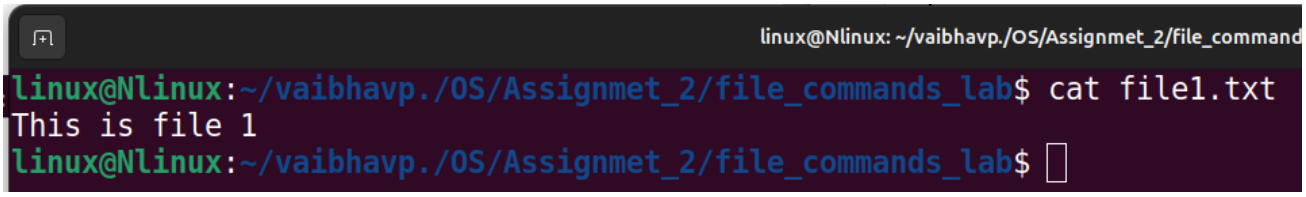


Question 15:

- **Use the cat command to display the contents of file1.txt.**

Commands:

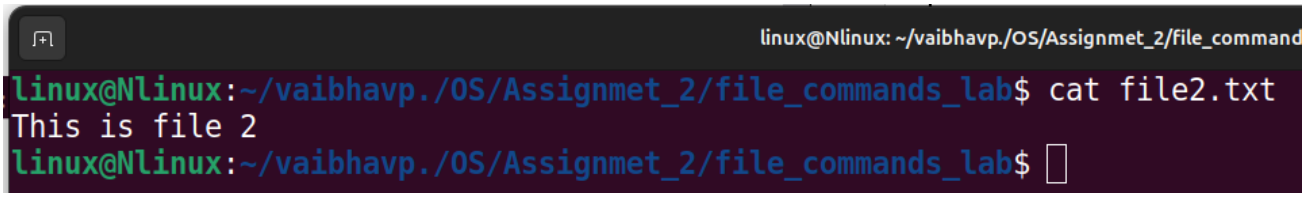
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ cat file1.txt



- **Use the less command to view the contents of 'file2.txt'**

Commands:

linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ cat file2.txt



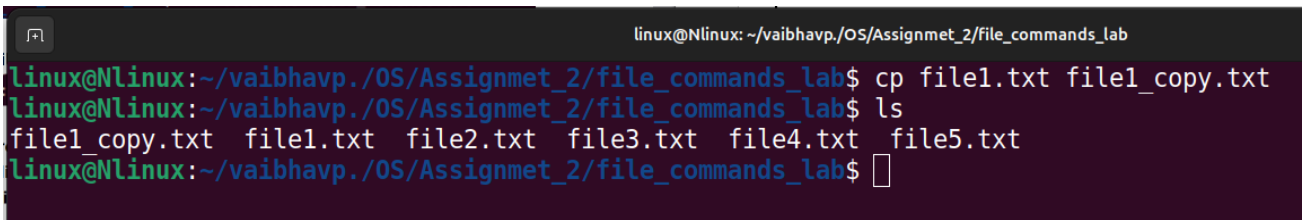
Question 16:

<u>Copy Files:</u>

- **Copy ‘file1.txt’ to create a new file named ‘file1_copy.txt’**

Commands:

linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ cp file1.txt file1_copy.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ ls

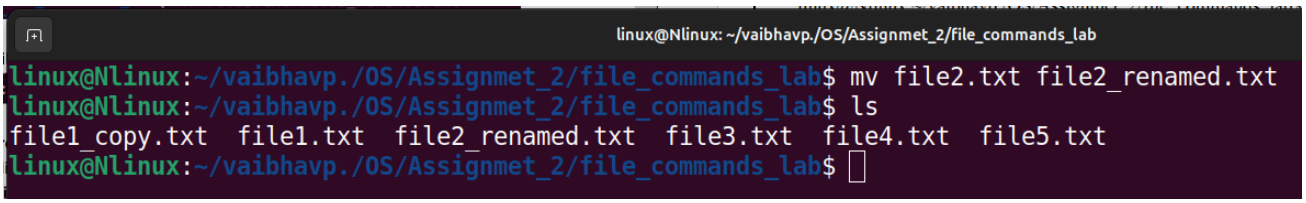


- **Rename ‘file2.txt ‘to ‘file2_renamed.txt’**

Commands:

linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ mv file2.txt file2_renamed.txt

linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ ls



- **Delete ‘file3.txt’.**

Commands:

linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ rm file3.txt

linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ ls

```
linux@Nlinux: ~/vaibhavp./OS/Assignmet_2/file_commar
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ rm file3.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ ls
file1_copy.txt  file1.txt  file2_renamed.txt  file4.txt  file5.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$
```

- **Change to the 'backup directory and list the files:**

Commands:

linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ mkdir backup

linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ mv file1_copy.txt file2_renamed.txt backup

linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ ls backup/

```
linux@Nlinux: ~/vaibhavp./OS/Assignmet_2/file_commands_lab
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ mkdir backup
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ mv file1_copy.txt file2_renamed.txt backup
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ ls backup/
file1_copy.txt  file2_renamed.txt
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$
```

Question 17:

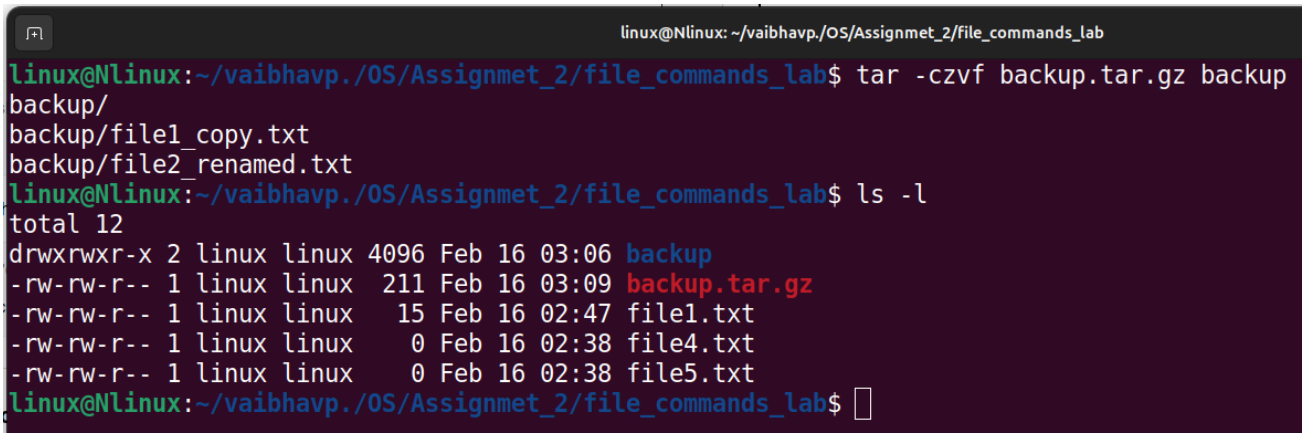
<u>Create a Compressed Archive:</u>

- **Go back to the 'file_commands_lab' directory and create a compressed archive of the backup directory.**

Commands:

linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ tar -czvf backup.tar.gz backup

linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ ls -l



Question 18:

<u>Extract the Archive:</u>

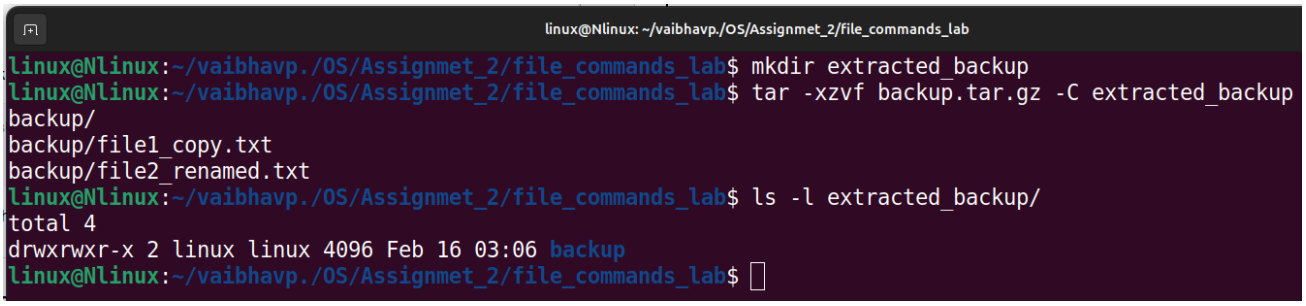
- **Create a new directory named 'extracted_backup' and extract the contents of 'backup.tar.gz' into it:**

Commands:

linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ mkdir extracted_backup

linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ tar -xzvf backup.tar.gz -C extracted_backup

linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ ls -l extracted_backup/



Question 19:

<u>Verify Extraction:</u>

- **List the contents of the 'extracted_backup' directory to verify that the files were extracted correctly.**

Commands:

linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ ls -l extracted_backup/

```
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ ls -l extracted_backup/
total 4
drwxrwxr-x 2 linux linux 4096 Feb 16 03:06 backup
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$
```

Question 20:

Create a text file named "count.txt" and write the 10 line sentences and use the command to count the word, lines, character and also use command for highlighting a patten in this file.

Commands:

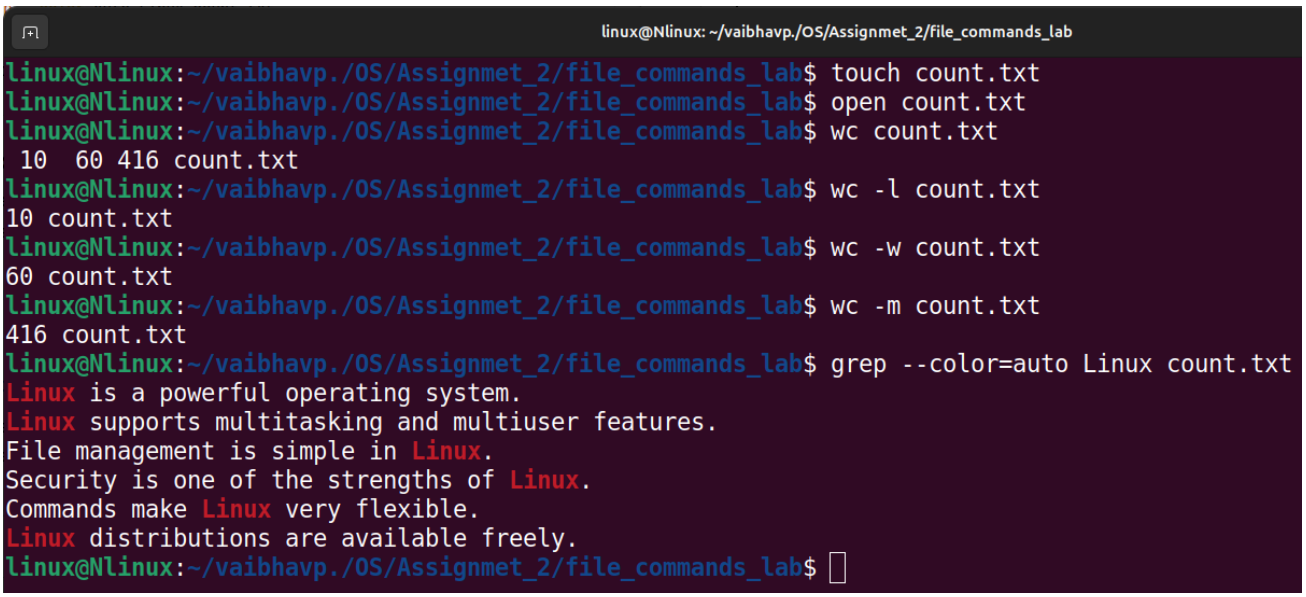
linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ touch count.txt

linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ open count.txt

linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ wc count.txt

linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ wc -l count.txt

linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ wc -w count.txt

linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ wc -m count.txt

linux@Nlinux:~/vaibhavp./OS/Assignmet_2/file_commands_lab$ grep --color=auto Linux count.txt